



Usage-Based Statistical Testing

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Usage-Based Statistical Testing

Usage-Based Statistical Testing focuses testing resources on software execution paths based on how users run the system in production, rather than focusing on structural code coverage.

Core Concept & Theory

- **User-Centric View:** Traditional testing checks code branches, but users only run specific paths. Usage-based testing prioritizes paths users visit most.
- **Operational Profile (OP):** A probability distribution mapping user operations to their run frequencies.
- **Reliability Estimation:** Testing based on the OP provides a realistic estimate of the system's Mean Time to Failure (MTTF) in production.

Checklist Limitations

- **Checklist Testing:** Uses static checklists of common bugs to audit features manually.
- **Limitations:** Checklists can become outdated, suffer from bias, and do not scale. They fail to test dynamic data interactions and edge cases.
- **Why Use OP instead:** Checklists test what testers remember. Operational Profiles test what users actually do, ensuring high reliability where it matters most.

Musa's Operational Profiles

John Musa developed a structured, step-by-step model to build operational profiles:

Five Steps to Build an OP

- 1. Find Customer Profile:** Identify user segments (e.g. admin, guest, subscriber).
- 2. Find User Profile:** Identify distinct systems or components users access.
- 3. Find System Modes:** Identify system states (e.g. startup, normal, peak load).
- 4. Find Functional Profile:** List functions the software performs.
- 5. Find Operational Profile:** Group functions into specific user operations. Assign probabilities based on log telemetry or marketing data.

Profile Representation Example

User Operation	Count / hr	Probability
Search Products	6,000	0.60 (60%)
Add to Cart	2,500	0.25 (25%)
Checkout Payment	1,000	0.10 (10%)
Update Profile	500	0.05 (5%)



Case Study: Cartridge Support Software

Applying Musa's operational profiling to printer driver utility software:

Project Overview & Goals

- **Target System:** PC utility software that manages ink cartridge statuses, prints nozzle test patterns, and calibrates printer alignment.
- **Context & Problem:** Limited verification window before software release. Standard code coverage checks were taking too long.
- **Objective:** Model a 4-step functional profile and assign probabilities based on printer telemetry logs, targeting testing on the most frequent print paths.

Cartridge Software Operational Profile

Operation Class	Telemetry Probability	Test Effort Allocation
Query Ink Levels	0.75 (75%)	High (Verify UI/cache)
Print Test Pattern	0.15 (15%)	Medium (Check drivers)
Nozzle Alignment	0.08 (8%)	Low (Basic validation)
Firmware Upgrade	0.02 (2%)	Critical (Risk focused)

Applications of Usage-Based Testing

How software engineering teams utilize usage testing and profiles in practice:

Telemetry-Driven Performance Tests

Web apps analyze production traffic logs to build operational profiles. Load tests replicate this profile distribution (e.g. simulating 80% reads and 20% writes) to verify database performance under realistic loads.

Medical Device Reliability Models

Infusion pumps and pacemaker test runs are designed based on patient usage modes (operational profiles), enabling accurate Mean Time to Failure reliability estimates for FDA approval.

Automotive Firmware Verification

Engine and ADAS controllers are tested using simulated driving profiles (city, highway, extreme conditions) to ensure safety in typical user environments.

Mobile App UI Layout Tests

App analytics track user clicks. Teams focus manual UI testing on highly visited pages, keeping lesser-visited settings pages on automated schedules.

THANK YOU

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